Muhammad Talha Khalid

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Mechanical engineer working where thermal-fluids and digital twins meet.

Research interests: thermal-fluids (two-phase heat transfer, CFD, fluid-structure interaction, microfluidic flow); physics-informed digital twins; liquid cooling for data centres and electronics.

EDUCATION

Bachelor of Engineering, Mechanical Engineering — Oct 2020 – Jun 2024
National University of Sciences and Technology (NUST), Islamabad

CGPA 3.70/4.00, top 5% of the cohort. Dean's Honours List in each of the last three semesters, for a top-ten CGPA.

Final year project: *Multi Axis Control Based Self Stabilizing Platform for Heavy Payloads*

RESEARCH EXPERIENCE

Research Assistant, Digital Twins for Data-Centre Cooling — Jun 2026 – Present
NUST, Islamabad · supervised by Prof Ibraheem Haneef — *Part-time*

- Writing a critical review of how close physics-informed twins are to real-time use in liquid-cooled data centres.
- Built a 184-paper corpus in 12 topic groups and compared 11 earlier reviews across 12 dimensions to locate the gap.
- Each twin is graded on two scales set out in the review: DT0 to DT6 for maturity, V0 to V5 for validation.
- Published twins, including ExaDigiT and DCVerse, end the cooling loop at a fixed cold-plate resistance with single-phase coolant. The review argues that boiling and conjugate heat transfer are the main missing physics.

Research Assistant, Thermal-Fluids (CFD and Multiphysics) — Mar 2026 – Present
NUST, Islamabad · supervised by Dr Emad Ud Din — *Part-time*

- **Tandem flexible flags (manuscript 4).** Ran 36 cases (a 6 × 6 grid of body-to-flag and flag-to-flag gaps) at Re = 300 on the group's in-house immersed-boundary FSI solver, extending results reported by Mujtaba et al. (2021). The study is a collaboration between Pakistani and South Korean universities.
- Wrote Python scripts to process about 440 GB of output. They trim the start-up transient, check each run for solver problems and batch Tecplot figures of drag, lift, energy and flag motion.
- **Microfluidic cell sorter (manuscript 3).** Built and validated a 2D COMSOL 6.4 model of a blood cell sorter, then ran 112 cases across four channel designs, four wall-roughness factors and seven Reynolds numbers.

Remote Research Assistant, Laser Surface Texturing of Bearings — Apr 2025 – Jun 2026
Qilu Institute of Technology, Jinan, China · supervised by Dr Jawad Aslam — *Part-time*

- Prepared the research proposal for laser-textured roller bearings (three texture shapes, 5–20% area density) and assisted with digital-twin frameworks for monitoring friction, wear and lubrication.

MANUSCRIPTS

1. Farzad, M. S., Haneef, I., **Khalid, M. T.**, Waris, R., Amin, F. and Khan, M. A. A. *Integrated Mechatronic Design of a Heavy-Payload Two-Axis Gimbal Under Cyclic Loading.* Manuscript in final revision.
2. **Khalid, M. T.** and Haneef, I. *Towards Physics-Informed Digital Twins for Data-Centre Liquid and Immersion Cooling Systems: A Review.* Manuscript in preparation.
3. Ud Din, E. and **Khalid, M. T.** *Effect of Channel Geometry, Wall Roughness and Reynolds Number on White and Red Blood Cell Separation in a Microfluidic Sorter.* Manuscript in preparation.
4. Ud Din, E. and **Khalid, M. T.** *Tandem Flexible Flags Behind an Inverted C-Shaped Body at Re = 300: Effect of Body-to-Flag and Flag-to-Flag Gaps.* Manuscript in preparation.

ENGINEERING PROJECTS

Multi Axis Control Based Self Stabilizing Platform for Heavy Payloads — Jun 2023 – May 2024
Final year project, team of four · project report · control system photos

- The team designed and built a two-axis (pitch and yaw) gimbal for heavy payloads. The frame was modelled in SOLIDWORKS, cast in aluminium and checked in ANSYS Mechanical for stress, deformation and fatigue.
- Responsible for the control subsystem. Developed the Arduino software for the 1,000 W BLDC servos, integrated MPU6050 feedback with a teammate's Kalman filter, tuned the PID loops and added fail-safes for the 40 A drivers.
- The follow-up manuscript (manuscript 1) tested the prototype with a 20 kg payload under a 200 N, 20 Hz cyclic load. It reports measured settling times of 0.38 s in yaw and 0.47 s in pitch, with overshoot of 1.2% or less.

Cooling-load estimate for a two-storey house — Feb – Jun 2023
HVAC course project, team of five · Carrier HAP · project report

- Estimated the cooling load of each conditioned room in a two-storey house in Lahore, first in Carrier HAP and then by hand. In five of the six rooms compared, the hand figure came out 9–12% above HAP. The kitchen, where the stove dominates, went the other way: 19% below (22,362 against 27,600 Btu/hr).

Lunar electric rover: structural design without FEA Sep 2022 – Jan 2023
Mechanics of Materials course project, team of five · SOLIDWORKS, MATLAB · final design report · CAD files

- Designed a 12-wheel rover and checked its structure by hand, as FEA was not allowed. The calculations, scripted in MATLAB, covered an indeterminate chassis frame, lunar thermal stresses, a 1 m drop, column buckling and shaft fatigue. They showed 6061 aluminium falling short, so the design moved to 7068 with a Ti-6Al-4V suspension.

Smaller projects 2021

- An interactive Python stress-strain model, where temperature and cooling rate shift the modulus and yield strength through simple assumed factors (source code). Also an RC car with a self-designed motor driver (photos).

PROFESSIONAL EXPERIENCE

Product Operations Engineer, OneScreen Solutions May 2025 – Present
Islamabad · interactive flat panels for customers in the US, Canada, Europe and the Middle East *Full-time*

- Wrote a Python panel-registration tool for the manufacturer's API; it saved 125+ hours on a 1,500+ panel rollout.
- Also wrote the scripts that now handle licence-key allocation, invoice capture from a desktop ERP with no API, and renewal and overdue-invoice reminders (Python, Google Apps Script, n8n).
- Documented 12+ internal workflows, which cut onboarding time for new staff by about 30%. Supported school-district rollouts in Duncanville and Lewisville, Texas, part of deployments worth over US$5 million.
- Support, training and customer success on US Eastern hours; supervise one junior team member.

Junior Wireline Engineer, Well Logging Energy (Pvt.) Ltd. Oct 2024 – Jan 2025
Field sites in Khyber Pakhtunkhwa *Full-time*

- Ran production logging (PLT) jobs at the Bettani, Togh and Nashpa wells, and set up a Power Automate flow for the tool inventory that cut retrieval time by about 20%.

Maintenance Engineering Intern, Attock Refinery Ltd. Jun – Jul 2024
Rawalpindi *Full-time*

- Four weeks across the refinery's maintenance sections and power plant. Assisted on pumps, compressors and heat exchangers, and took part in root cause analyses that used fault tree analysis and FMEA.

Engineering Intern, Sui Northern Gas Pipelines Ltd. (SNGPL) Aug – Sep 2022
Islamabad *Full-time*

- Most of the internship was spent with Corrosion Control on pipeline inspection, corrosion prevention and leak detection. Also visited compression stations and observed how SCADA monitors more than 9,000 km of pipeline.

Mathematics Solver, Photomath Dec 2021 – Jun 2022
Remote *Freelance*

- Solved and reviewed 100+ student problems, writing step-by-step solutions in LaTeX with GeoGebra figures.

LEADERSHIP AND SERVICE

Mentor Technical, ASHRAE NUST Student Branch Sep 2023 – May 2024
Treasurer of the same branch, Sep 2022 – Sep 2023

- Ran the branch's technical workshops, mentored junior members and secured US$5,400 in funding for their training. As treasurer, kept the budgets and records with the university finance office.

Volunteer, Islamabad schools and Khalq Foundation Jan – Jun 2023

- Raised funds for two drinking-water coolers at under-resourced schools near D-11, Islamabad, and taught the students Excel. Also helped run a clothing drive for flood-hit families (record).

HONOURS

Prime Minister's Merit Laptop Scheme (2023). Third place, AKDC Design Challenge, NUST. First place, Stars of Chakwal, a district-level contest (2018).

SKILLS AND LANGUAGES

Simulation: Immersed-boundary CFD (in-house Fortran), COMSOL (laminar flow, particle tracing), ANSYS Mechanical, Tecplot, Carrier HAP, MATLAB
Design and hardware: SOLIDWORKS, AutoCAD, Arduino, LabVIEW, PID control
Data and automation: Python, C++, Google Apps Script, REST APIs, n8n, Zapier, Power Automate, LaTeX
Languages: English (IELTS Academic 8.0, August 2026); Urdu (native)

REFEREES

Prof Ibraheem Haneef
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Dr Emad Ud Din
Former Head of Department, Mechanical Engineering, SMME, NUST · emaduddin@smme.nust.edu.pk

Dr Jawad Aslam
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